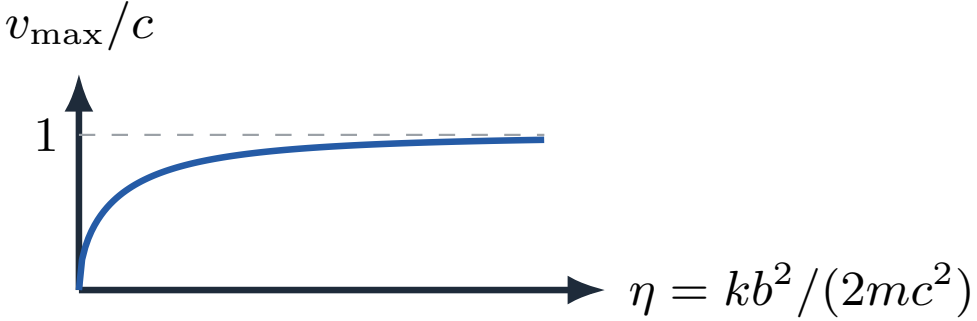


$$v_{\max}/c = \sqrt{1-(1+\eta)^{-2}}$$

$$T = 2\pi\sqrt{m/k}\,[1+3\eta/8+O(\eta^2)]$$

一阶周期公式只用于 $\eta \ll 1$